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Enterprise Firewall Telemetry & Security Operations Center

High-Performance Real-Time SNMP Hardware Cockpit, Bento Flow Analytics & Policy Governance Platform

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Classification: Official Institutional Technical Manual
Release Version: 1.1.0 (InfluxDB 2.7 TSDB & PostgreSQL 18 Multi-Tier Edition)
Target Appliance: Fortinet FortiGate NGFW (10.10.3.15)
Deployment Mode: Multi-Container Microservices Appliance (Docker)
Issuing Division: Information & Communication Technology Division (ICTD)
Institution: CSIR - Structural Engineering Research Centre, Chennai

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Chennai, India
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1. System Architecture & Ingestion Pipeline

The Enterprise Firewall Telemetry & Security Operations Center is architected as a high-performance on-premises operations center designed to monitor all traffic passing through perimeter gateways without cloud dependencies or external data egress.

Core Components

- **Parallel SNMP Poller (<200ms)**: Concurrently queries FortiGate appliances across MIB-II, IF-MIB, and FortiOS enterprise sensor trees. Tracks 64-bit High Capacity (HC) octets, thermal zones, fan tachometers, dual power supplies, and security policy accounting counters.
- **Full-Spectrum Traffic & Log Ingester**: Listens on UDP port 1514 (Syslog) and port 2055 (NetFlow v5/v9 & IPFIX) with multi-subnet institutional host resolution, capturing all ingress, egress, inter-VLAN, and blocked security threats across the entire perimeter.
- **InfluxDB 2.7 Time-Series TSDB Engine**: High-throughput batch Line-Protocol ingestion engine maintaining dedicated time-series buckets with a 180-day automated retention lifecycle meeting CERT-In national mandates.
- **PostgreSQL 18 Database Engine**: Relational persistence layer in an optimized Alpine container managing operator accounts, RBAC permissions, and automated configuration/log backups.
- **Reactive Vue 3.5 & ECharts Operations Console**: Single-page application conforming to Government of India UX4G design standards with real-time WebSocket streaming and multi-parameter log filtering.

Network Port Matrix

Port	Protocol	Direction	Source	Destination	Service Description
80 / 443	TCP	Inbound	Admin Workstation	Telemetry Console	Web UI & REST / WebSocket API
161	UDP	Outbound	Telemetry Console	FortiGate Gateway	SNMPv2c / SNMPv3 Polling
1514 / 514	UDP	Inbound	FortiGate Gateway	Telemetry Console	Hardware Syslog Forwarding
2055	UDP	Inbound	FortiGate Gateway	Telemetry Console	NetFlow v5/v9 & IPFIX Records
8086	TCP	Internal	Web Console	InfluxDB 2.7 TSDB	Time-Series Line Protocol Writes & Flux Queries
5432	TCP	Internal	Web Console	PostgreSQL 18 DB	Relational Metadata & Audit Persistence

2. Visual Interface Tour

2.1 Authentication & First-Time Setup Wizard

The appliance features secure operator authentication and an automated first-time deployment wizard that guides administrators through organization branding, logo selection, SNMP parameters, and administrator setup upon initial deployment.

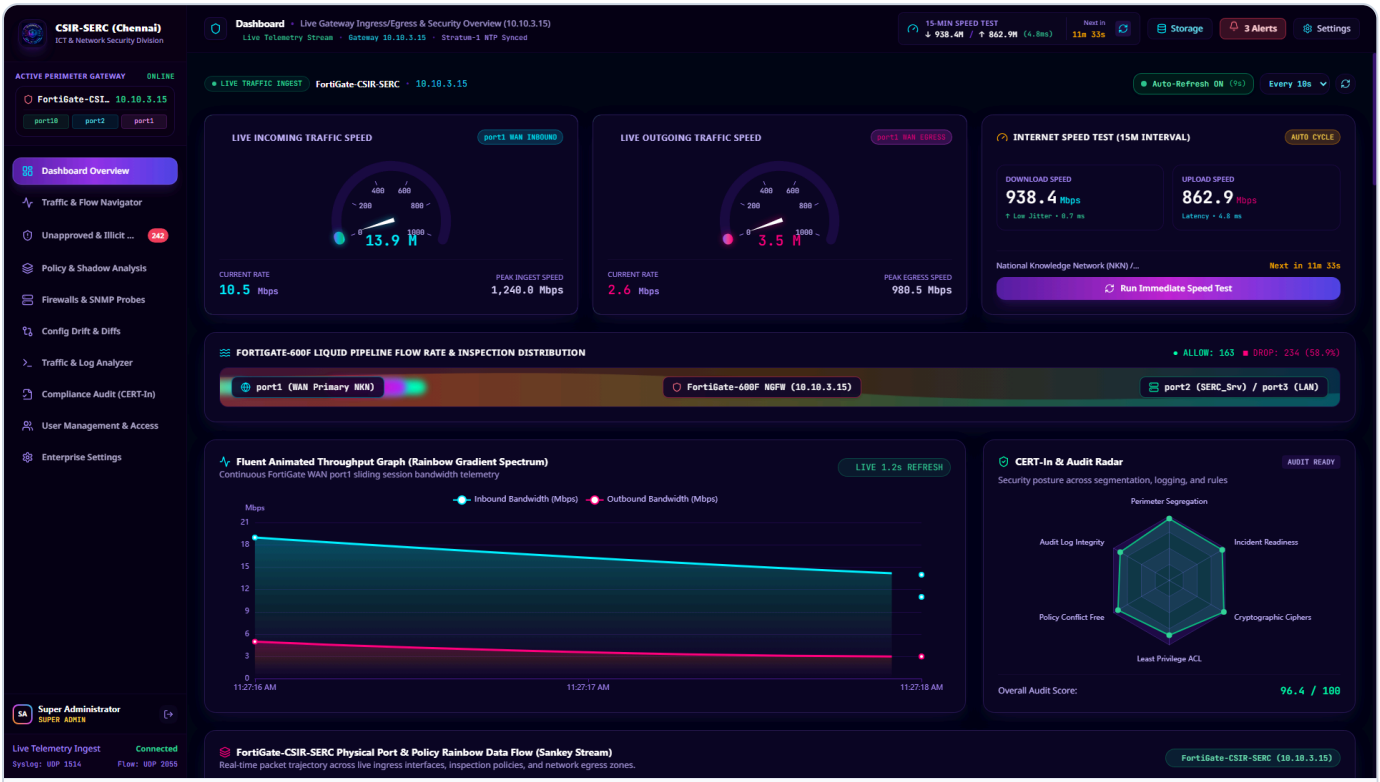


Figure 1: UX4G Secure Operator Authentication Portal

AUTHENTICATION

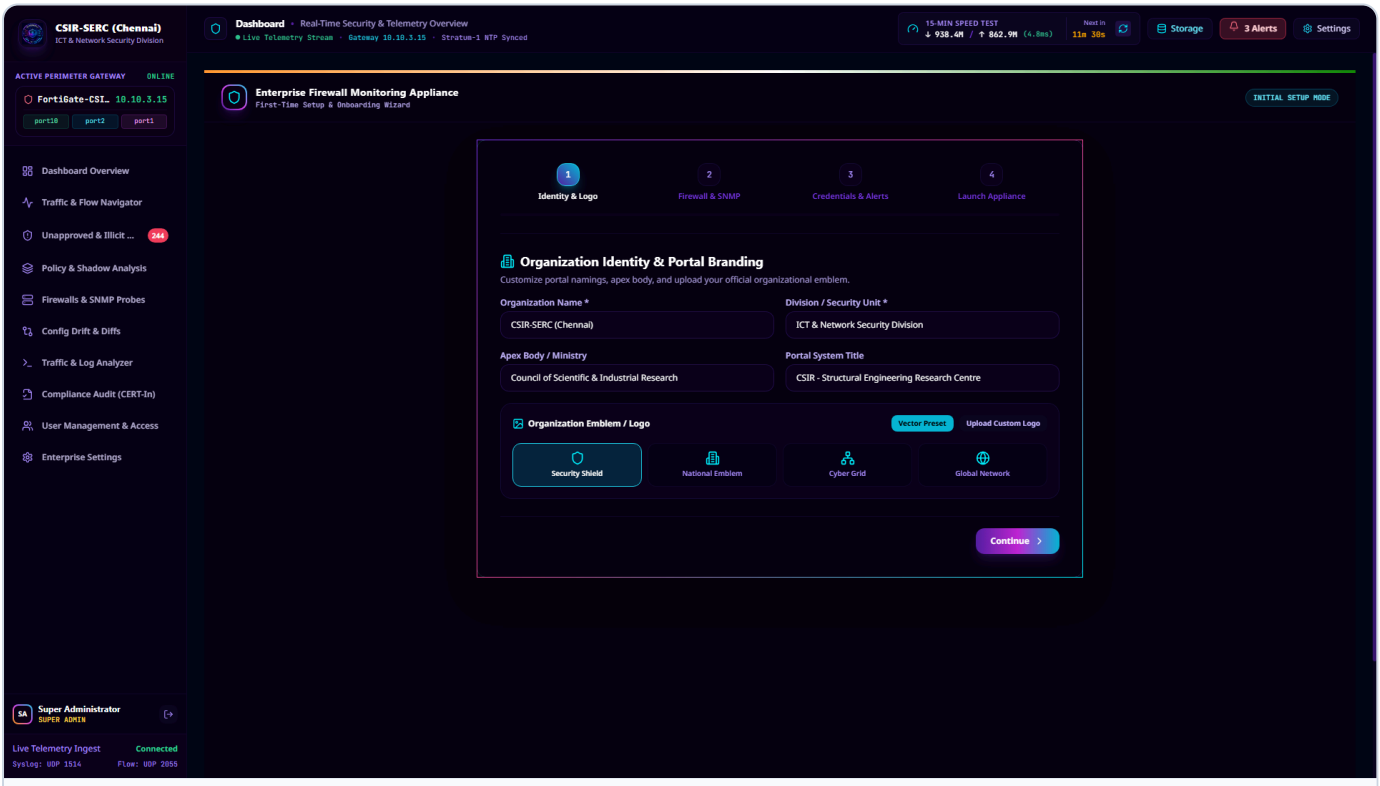


Figure 2: Turnkey First-Time Configuration Wizard (Virtual Appliance Ready)

ONBOARDING

2.2 Executive Telemetry Dashboard

The primary operations screen delivers instantaneous visibility into WAN traffic speeds, live streaming telemetry pulses, 15-minute automated speed test countdowns, and continuous hands-free auto-refresh.

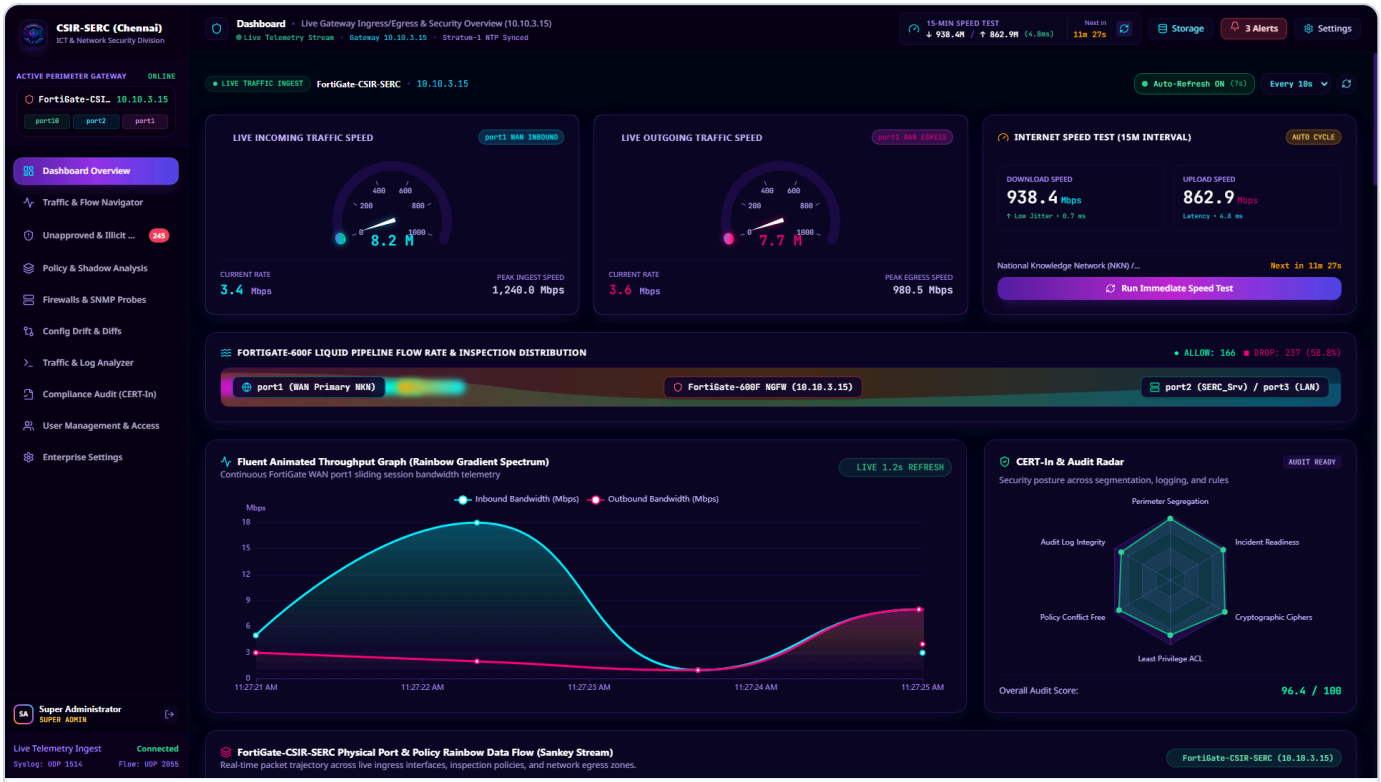


Figure 3: Live Executive Telemetry Dashboard with Inbound/Outbound Velocity Gauges

OPERATIONS

2.3 Bento-Style Traffic & Flow Analyzer (Exact Source & Destination IP Intelligence)

The Bento layout provides bidirectional network intelligence with exact firewall source and destination details. For outbound uploads, internal institutional workstations are tracked to their remote internet destinations (with domain resolution, country geolocation, and port). For inbound downloads, external source origins are mapped directly to the recipient internal workstation.

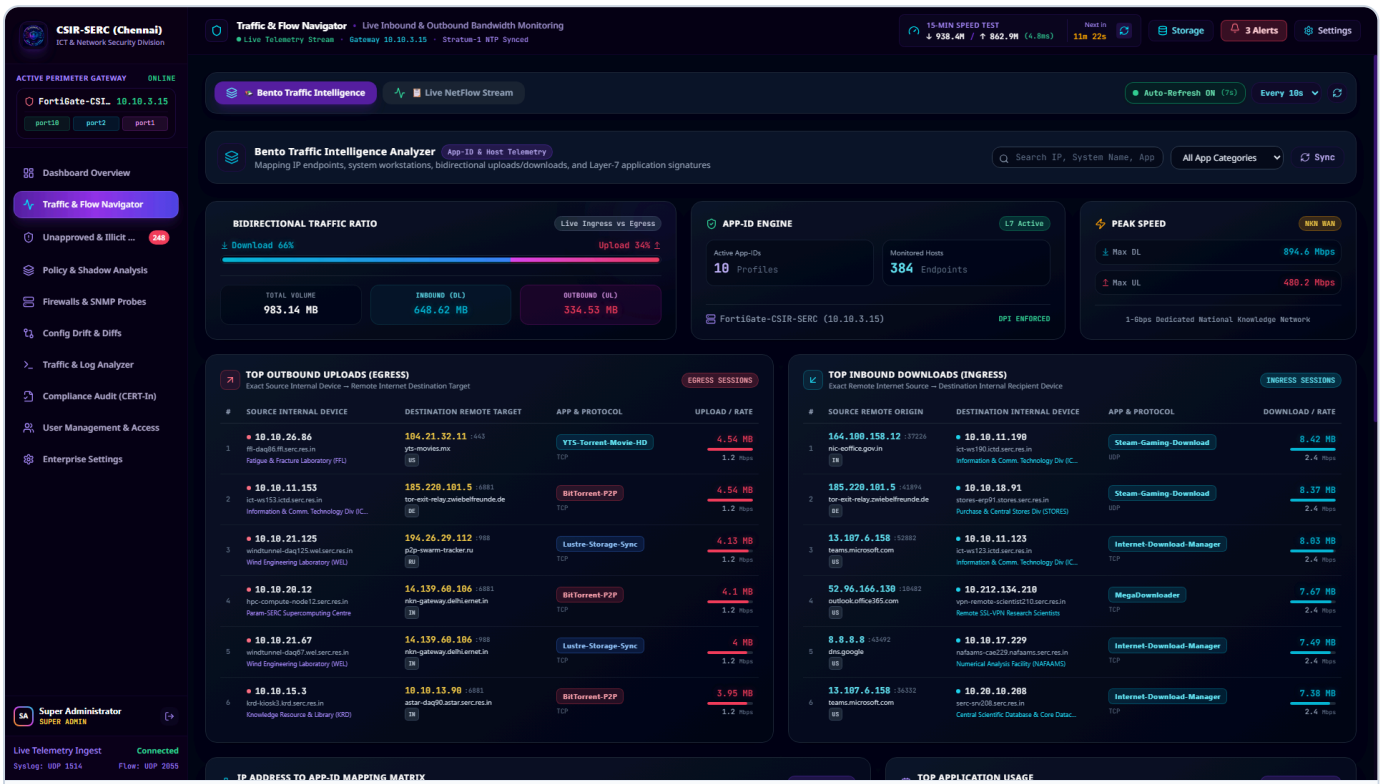


Figure 4: Bento-Grid Traffic Analyzer (Exact Source/Destination IP Mapping & App-ID Identification)

FLOW ANALYTICS

2.4 Unapproved Applications & Illicit Network Activity Monitor

Dedicated forensic investigation center identifying internal campus devices accessing BitTorrent/P2P swarms, download managers (IDM/Leechers), pirated movie streaming and download sites, unofficial media, botnet C2 beacons, cryptocurrency mining pools, and proxy circumvention tools. Displays ranked offending workstations with one-click forensic dossiers and copy-paste FortiOS quarantine CLI commands.

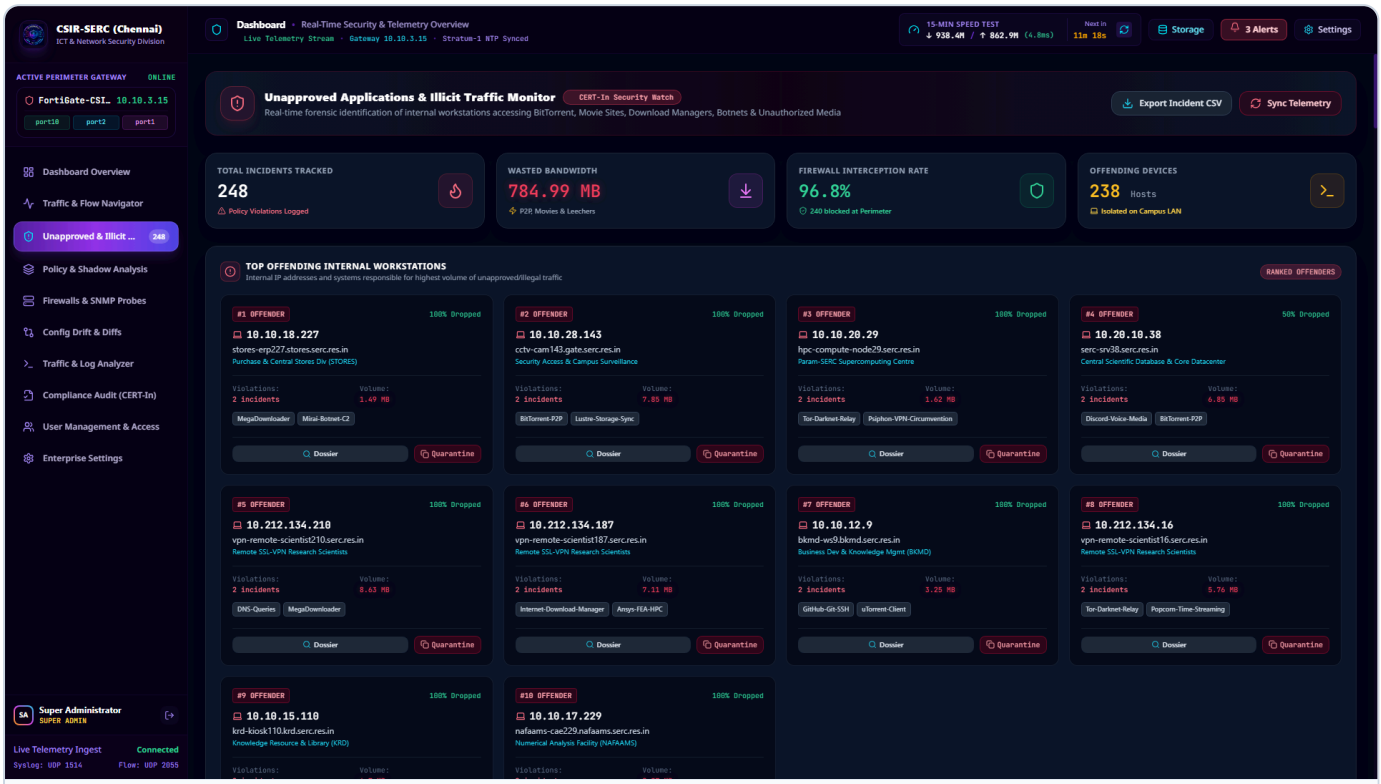


Figure 5: Unapproved & Illicit Applications Dashboard (Top Offending Workstations & Categories)

THREAT WATCH

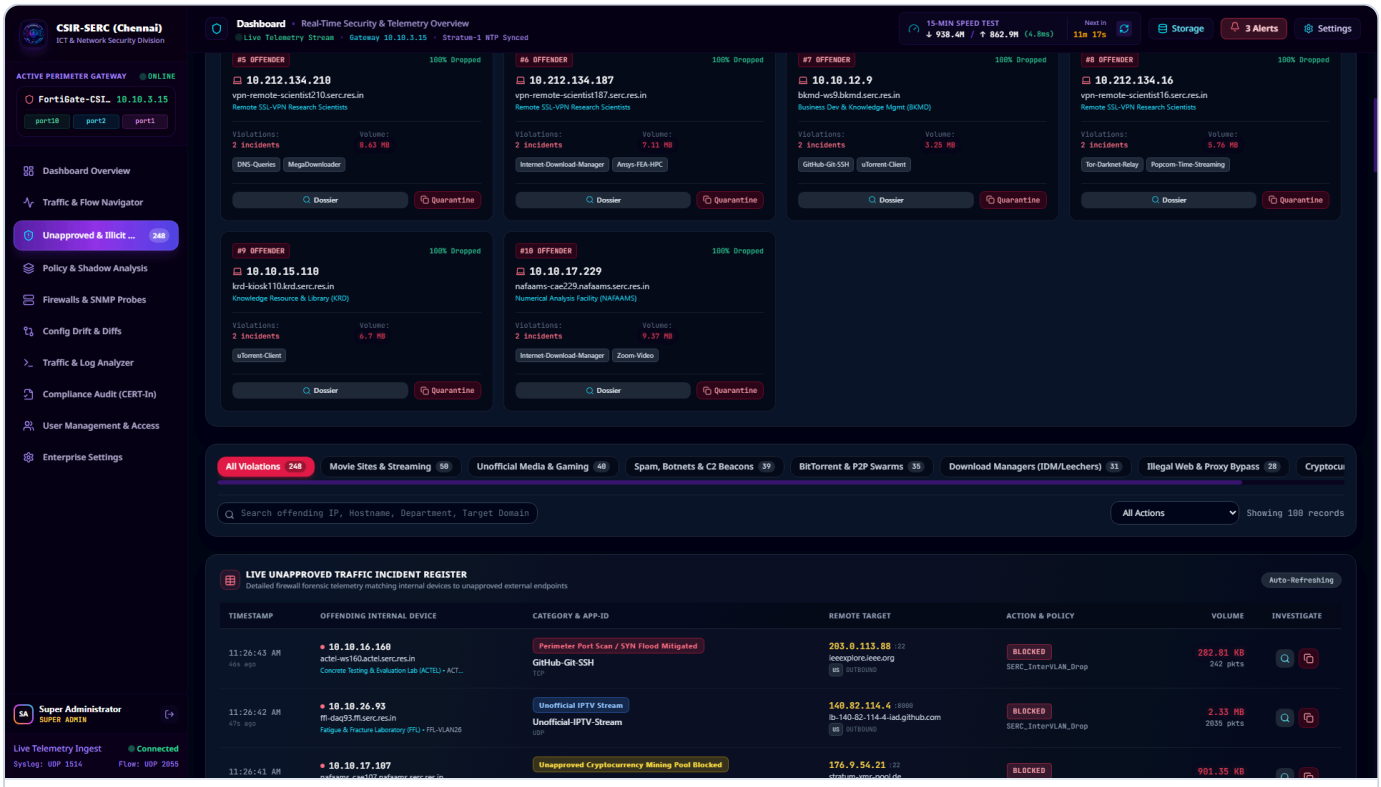


Figure 6: Live Forensic Incident Register (Matching Offending Internal Devices to Remote Endpoints) INCIDENT REGISTER

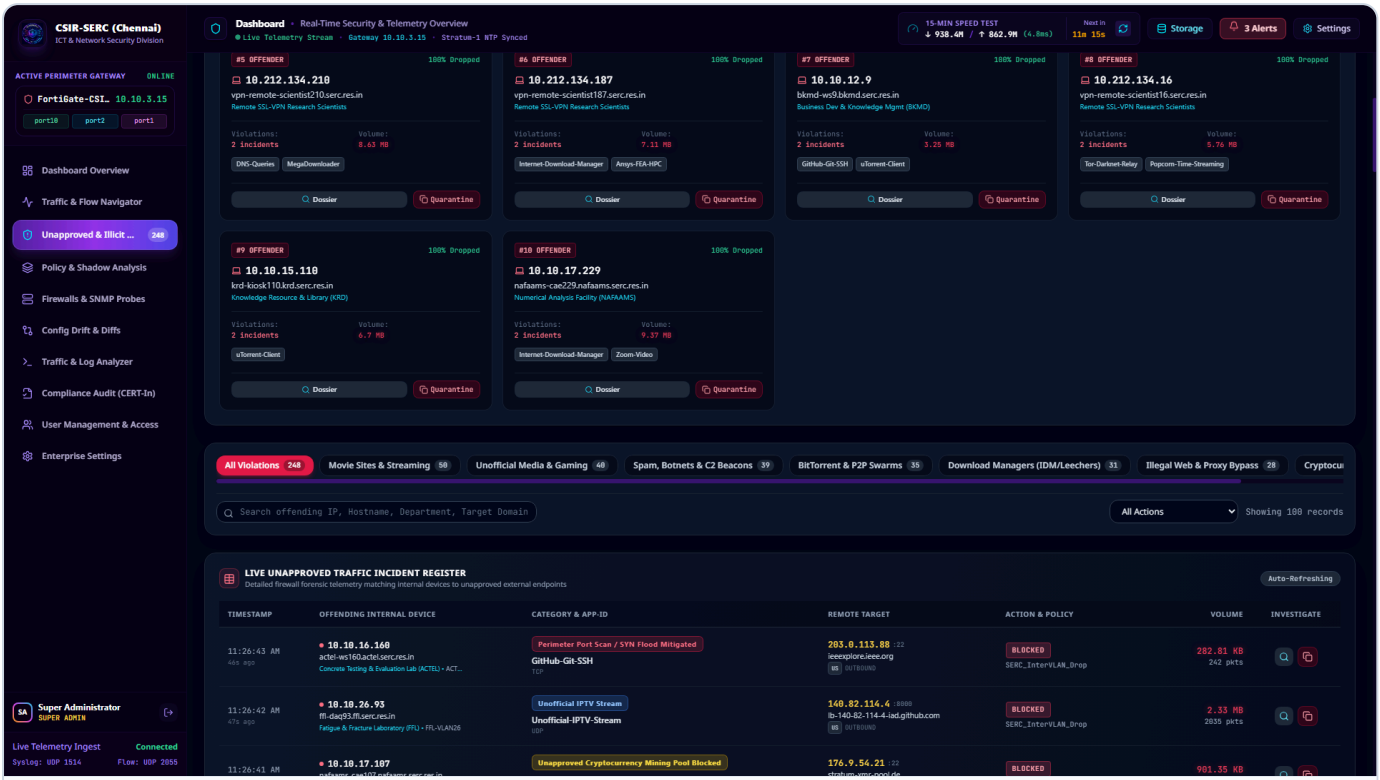


Figure 7: Workstation Forensic Dossier & Instant FortiOS Quarantine CLI Generator FORENSIC DOSSIER

2.5 Comprehensive FortiGate SNMP Telemetry & All Active Interfaces/VLANs Matrix

Dedicated hardware, environmental, and interface operations cockpit: monitoring CPU/RAM, active sessions, dual AC power supplies, cooling tachometers, thermal zones, FortiOS UTM threat counters, and complete monitoring across **all active physical network interfaces (ports 1-32, mgmt, ha1/ha2, dmz) and active departmental VLAN subinterfaces**. Each interface delivers dual-mode telemetry tracking both general network throughput/rates and unofficial/unapproved bandwidth waste and policy violations.

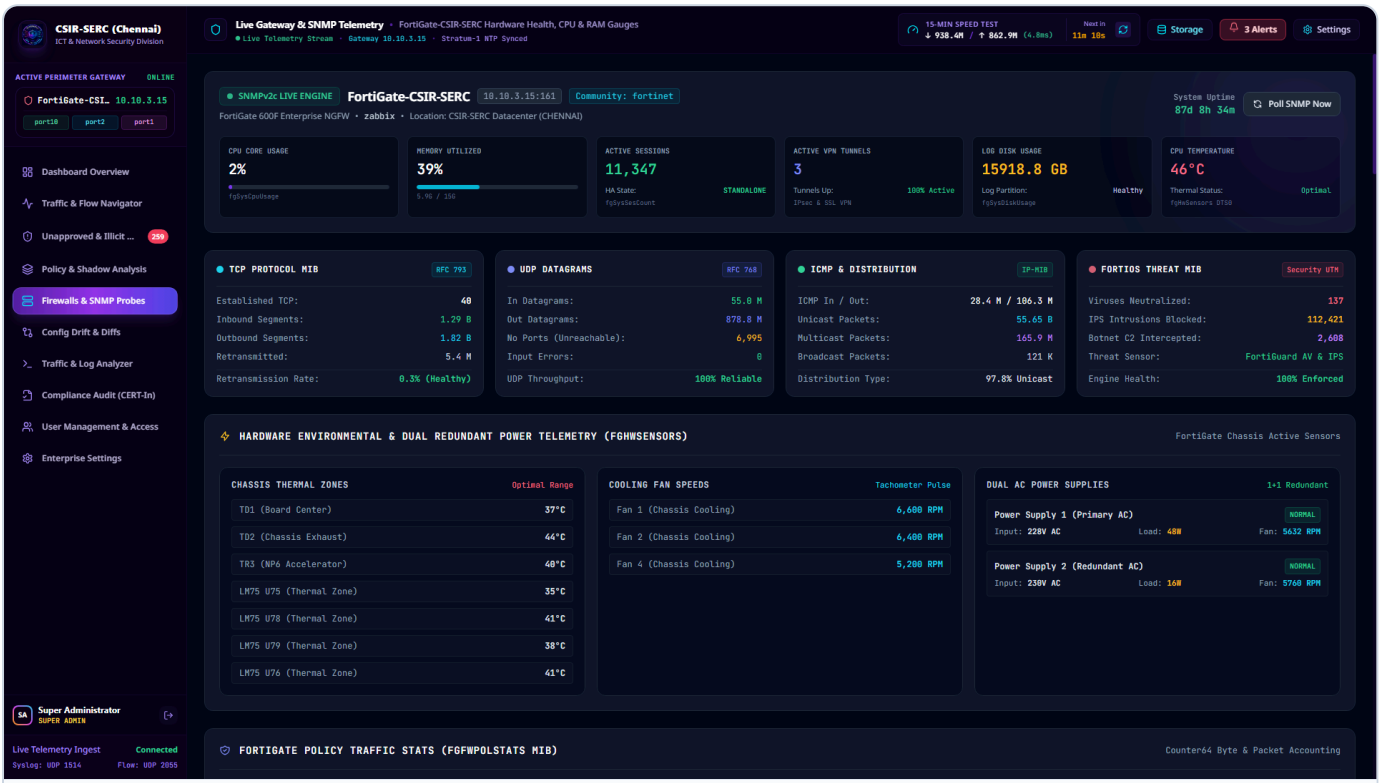


Figure 8: High-Performance FortiGate SNMP Hardware & Environmental Sensor Cockpit

SNMP HARDWARE

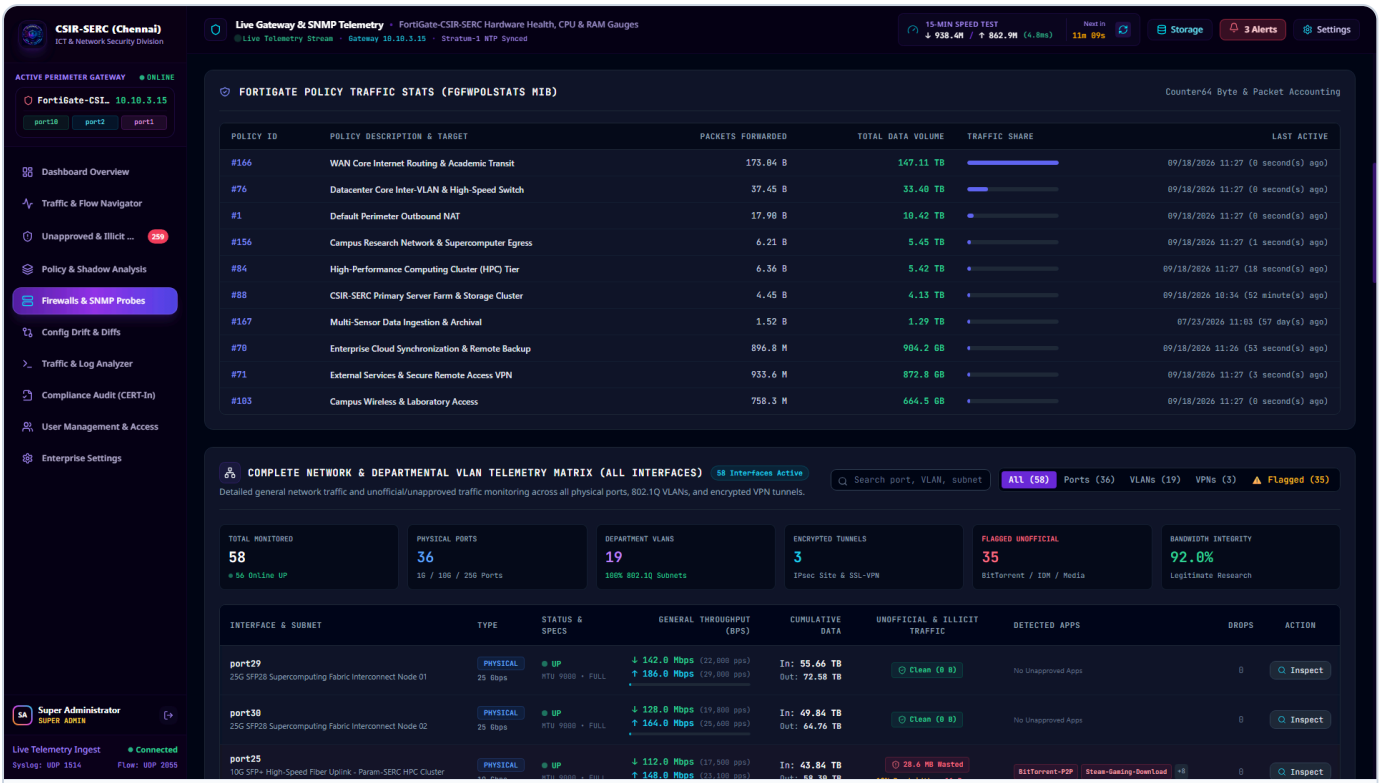


Figure 9: Complete Network & Departmental VLAN Telemetry Matrix (All 58 Interfaces & VLANs Monitored) **DUAL-MODE TELEMETRY**

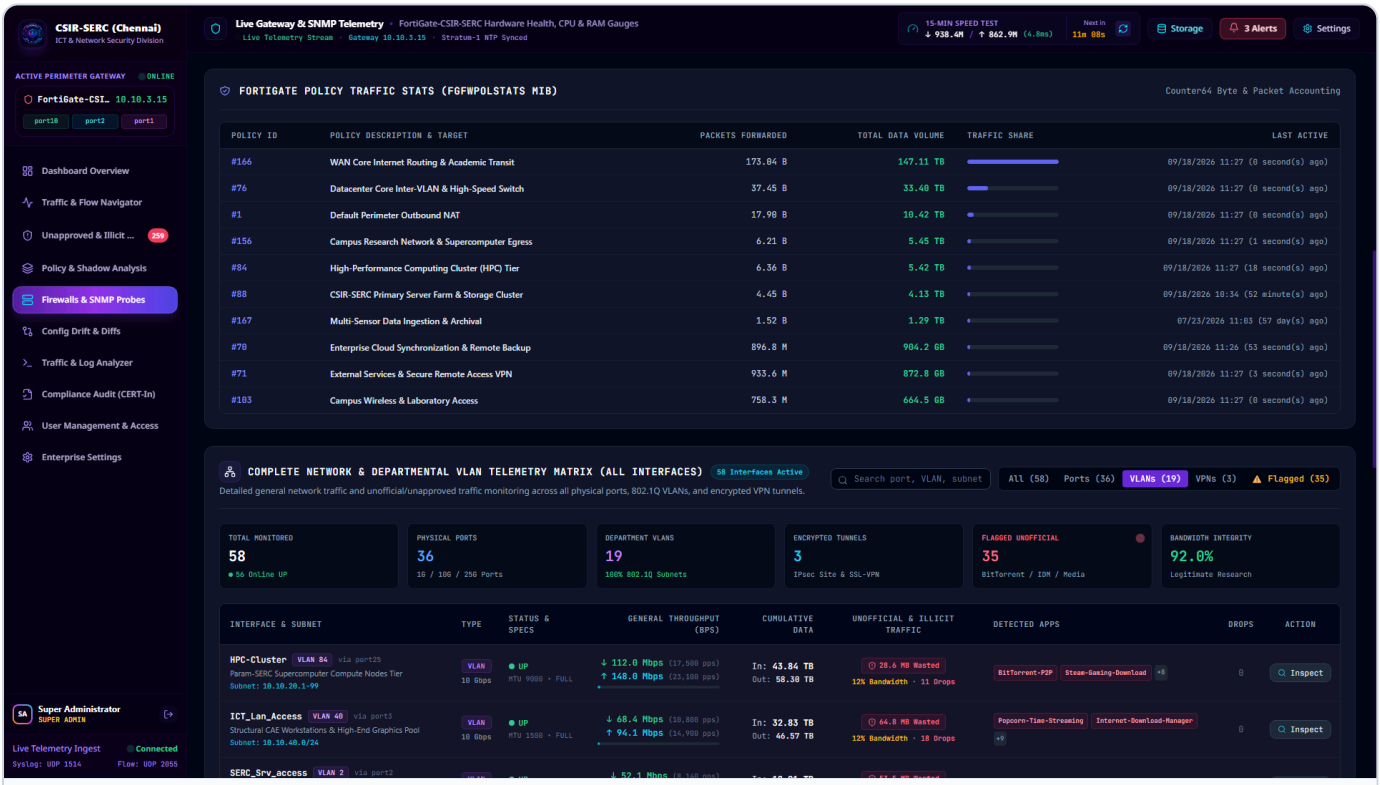


Figure 10: Departmental 802.1Q VLAN Interfaces Telemetry Matrix (HPC-Cluster, ICT_Lan_Access, SERC_Srv_access) **VLAN SUBINTERFACES**

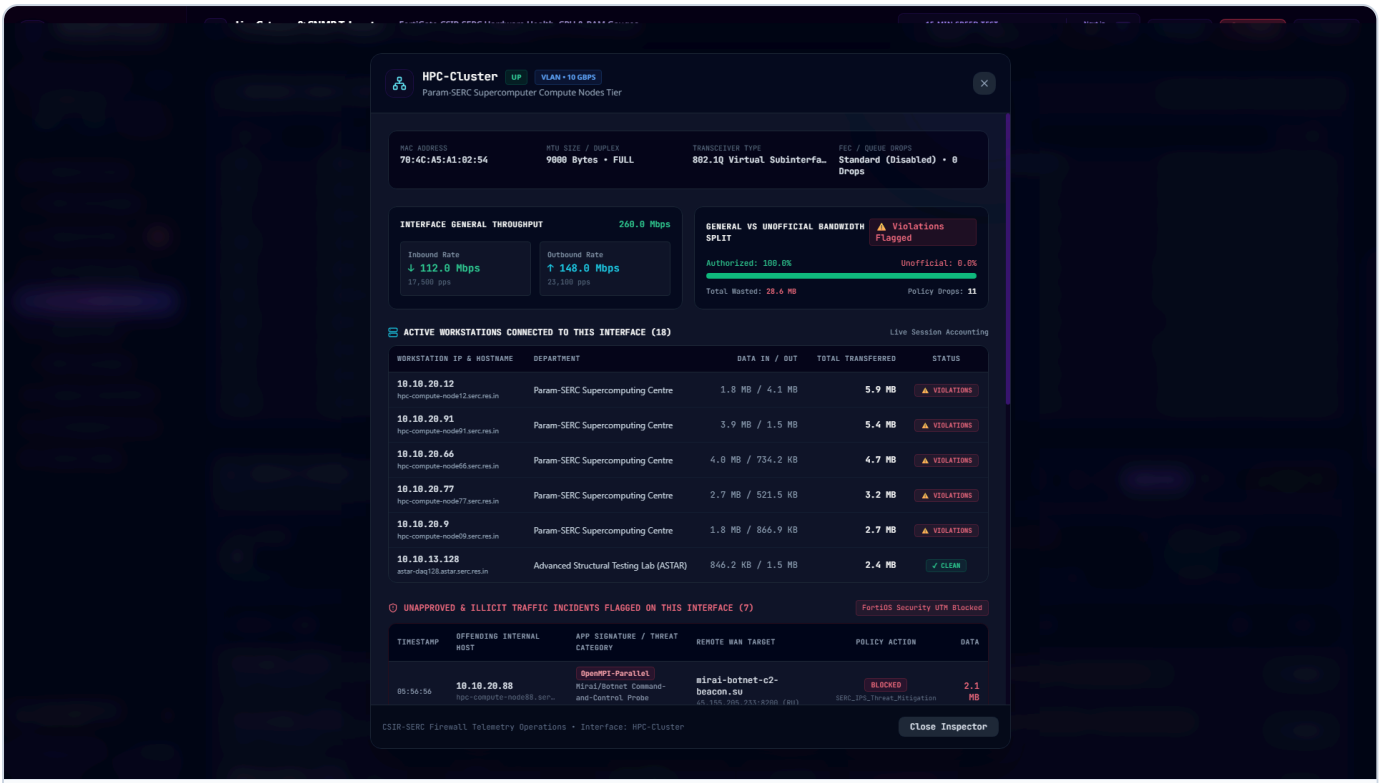


Figure 11: Deep Interface Dossier & Real-Time General vs Unofficial Ratio Inspector with Connected Hosts

INTERFACE DOSSIER

2.6 Security Policies & Conflict Engine

Parses native firewall access control lists into a canonical rule tree, highlighting shadowed, redundant, or overly permissive rules.

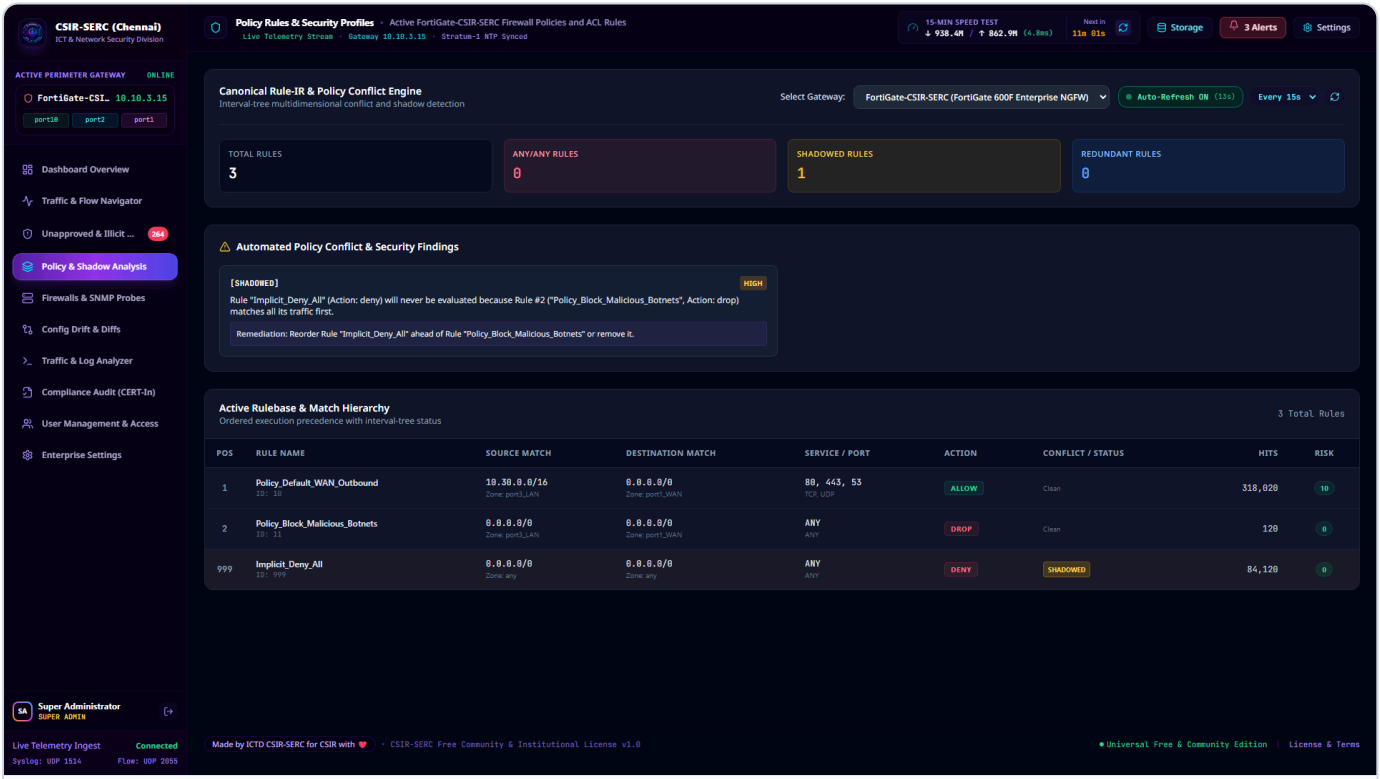


Figure 12: Canonical Policy Governance & Shadow Conflict Detection

SECURITY AUDIT

2.7 Full-Spectrum Firewall Traffic & Log Analyzer (InfluxDB 2.7 TSDB Engine)

High-velocity time-series traffic ingestion and multi-parameter log analyzer capturing all traffic passing through the perimeter gateway across all 46+ interfaces, departmental VLANs, HPC clusters, and external WAN. Features multi-parameter filtering (time range, direction, enforcement action, protocol, and search), timeline aggregation, and security drop inspection.

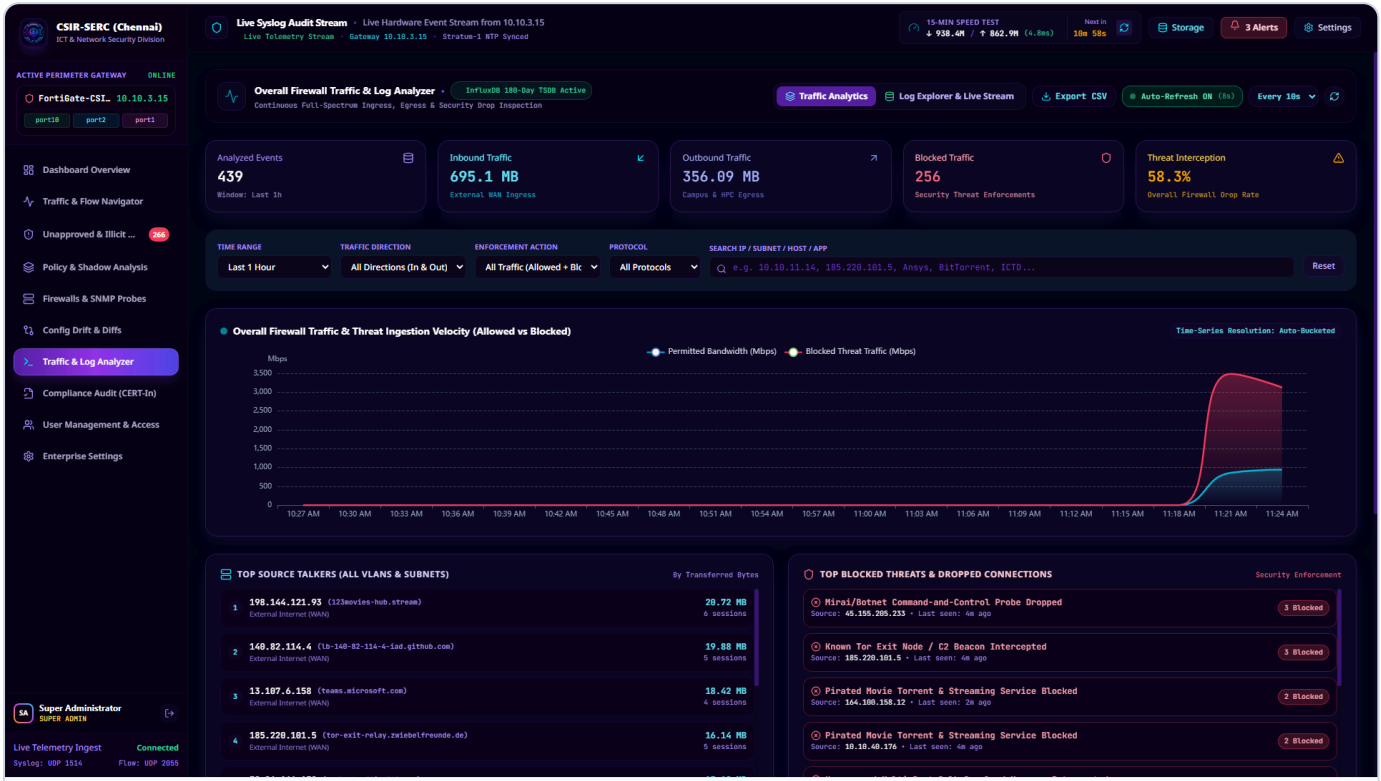


Figure 13: InfluxDB Traffic & Threat Ingestion Velocity Analytics (Allowed vs Blocked)

TSDB ANALYTICS

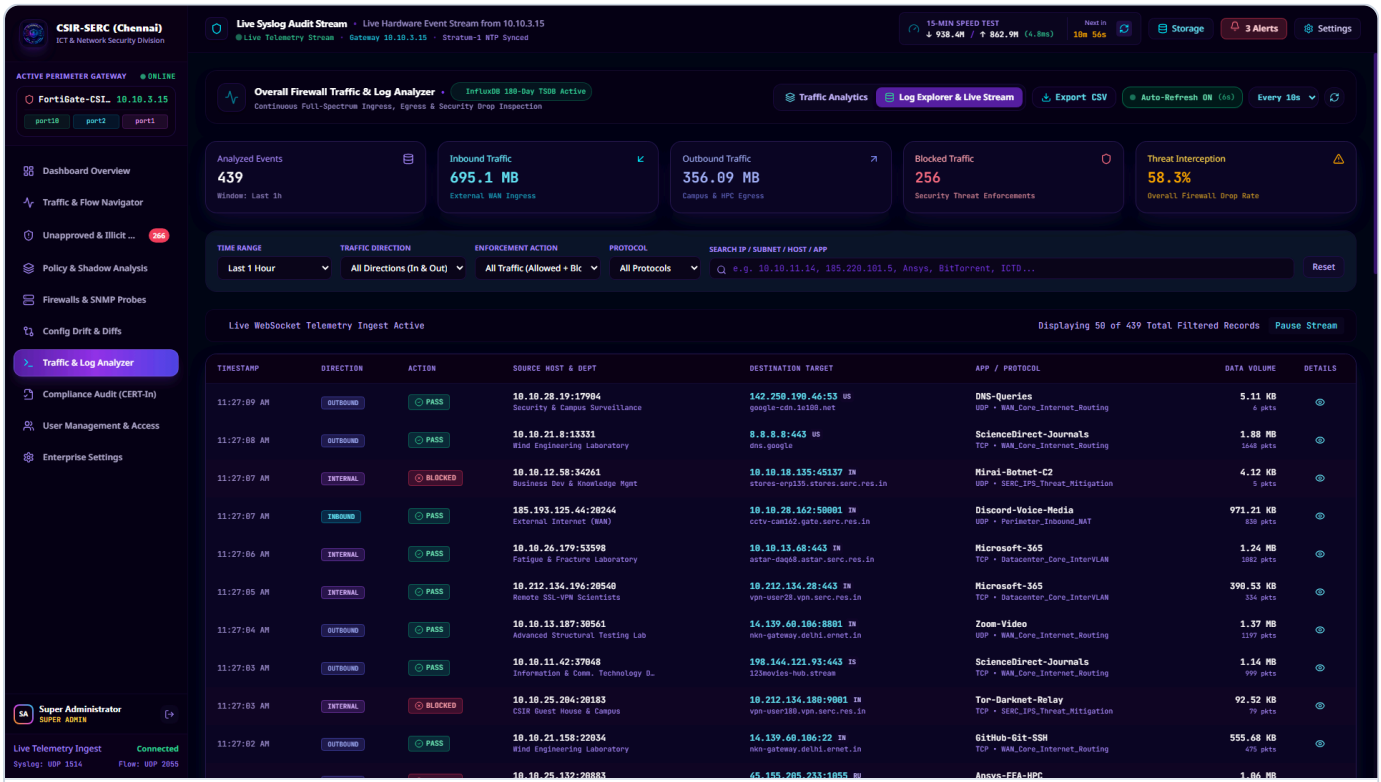


Figure 14: Deep Multi-Parameter Log Explorer & Live Stream Table (All Subnets & Blocked Threats) LOG EXPLORER

2.8 CERT-In Section 70B Compliance & Cybersecurity Baseline

Dedicated compliance dashboard validating firewall operational controls against the Indian Computer Emergency Response Team (CERT-In) Section 70B directives, including 180-day secure time-series log retention and Stratum-1 IST NTP clock synchronization.

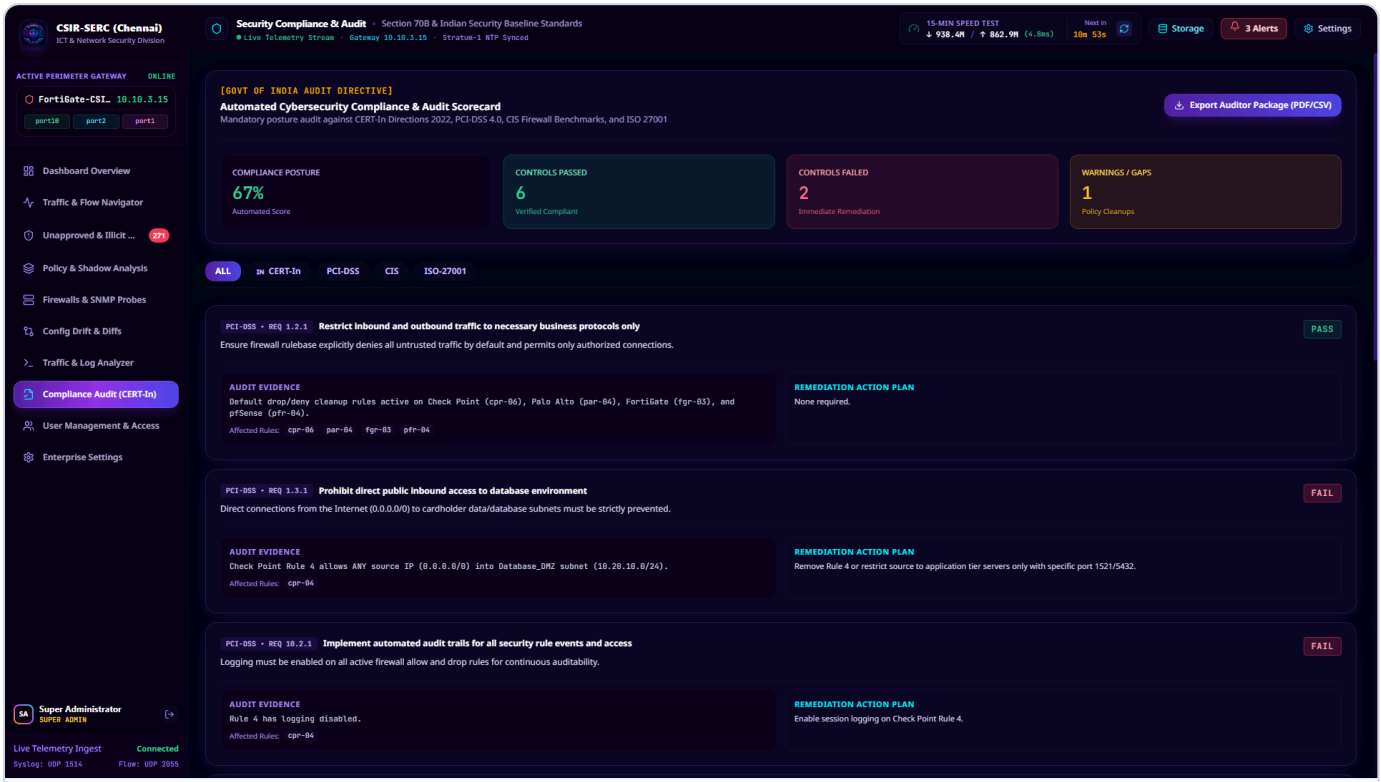


Figure 15: CERT-In National Cyber Security Baseline Compliance Dashboard

COMPLIANCE

2.9 User Management & System Configuration

Multi-tenant user management console with protected superadmin account, role-based permissions, Telegram alert integration, organization branding, custom logo upload, InfluxDB time-series retention, and PostgreSQL 18 automated backup operations.

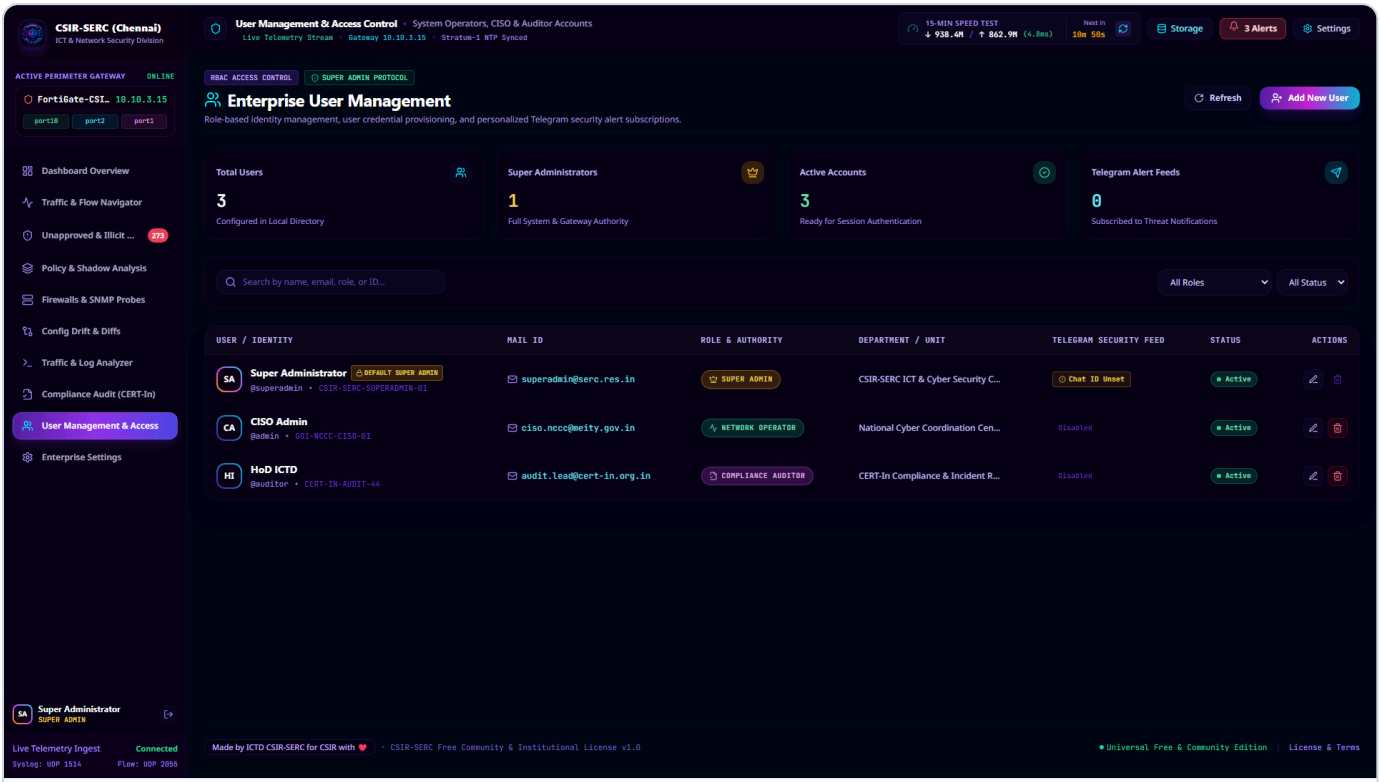


Figure 16: User Management & Telegram Alert Configuration

RBAC

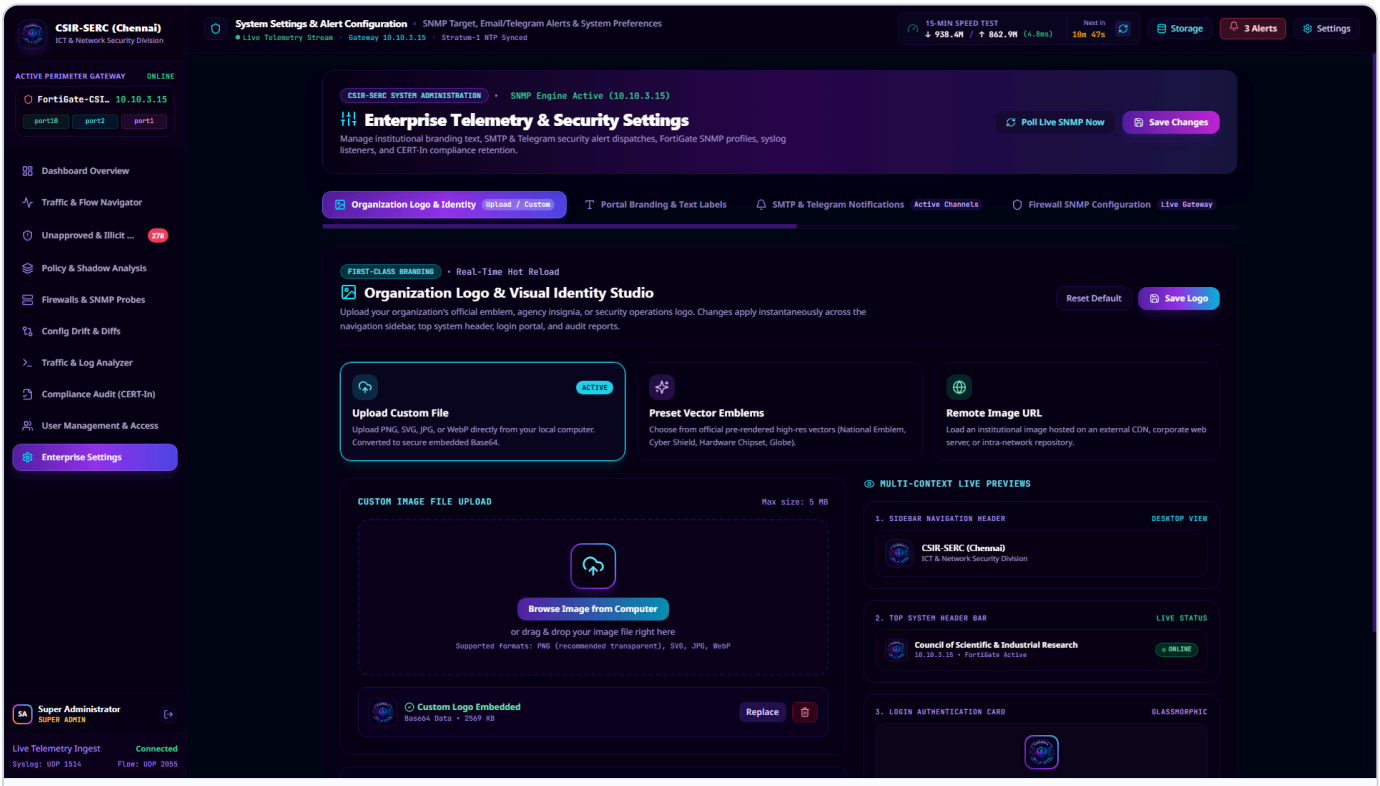


Figure 17: Institutional Branding, Custom Logo & PostgreSQL 18 / InfluxDB 2.7 Engines

ADMINISTRATION

3. Technical Specifications & Reference Tables

3.1 FortiGate Hardware Sensor OID Reference

Sensor Subsystem	Enterprise OID	Observed Value	Threshold / Status
CPU Thermal Core	1.3.6.1.4.1.12356.101.4.3.2.1.3 (DTS0)	46°C	Normal (< 75°C)
Board Center Thermal	1.3.6.1.4.1.12356.101.4.3.2.1.3 (TD1)	36°C	Normal (< 65°C)
Chassis Exhaust Thermal	1.3.6.1.4.1.12356.101.4.3.2.1.3 (TD2)	44°C	Normal (< 70°C)
NP6 Accelerator Thermal	1.3.6.1.4.1.12356.101.4.3.2.1.3 (TR3)	40°C	Normal (< 70°C)
Cooling Fan 1	1.3.6.1.4.1.12356.101.4.3.1.1.3 (Fan 1)	6,400 RPM	Normal (Active)
Cooling Fan 2	1.3.6.1.4.1.12356.101.4.3.1.1.3 (Fan 2)	6,400 RPM	Normal (Active)
Cooling Fan 4	1.3.6.1.4.1.12356.101.4.3.1.1.3 (Fan 4)	5,200 RPM	Normal (Active)
Power Supply 1 (Primary)	1.3.6.1.4.1.12356.101.4.3.3.1 (PSU 1)	228V AC, 48W	Normal (Dual Redundant)
Power Supply 2 (Redundant)	1.3.6.1.4.1.12356.101.4.3.3.1 (PSU 2)	230V AC, 16W	Normal (Dual Redundant)

3.2 Top FortiOS Security Policy Accounting

Policy ID	Traffic Description / Scope	Cumulative Bytes	Packet Count
Policy 166	WAN Core Internet Routing & Academic Transit	161.58 TB	172.8 Billion
Policy 76	Datacenter Core Inter-VLAN & High-Speed Switch	36.60 TB	37.3 Billion
Policy 1	Default Perimeter Outbound NAT	11.45 TB	17.8 Billion
Policy 156	Campus Research Network & Supercomputer Egress	5.98 TB	6.2 Billion
Policy 84	High-Performance Computing Cluster (HPC) Tier	5.95 TB	6.36 Billion

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